MAT 443 Data Mining and Statistical Learning
Final Exam

Name: ______________________________

This is a take-home final exam. You are not allowed to discuss it with your classmates. You are required to submit your code through Canvas. If you fail to submit your code or exam by the deadline, you will get 20% off your score. Any work later than 2 hours will not be accepted!

Data Description:

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1. Title: Malicious network flow detection

2. Source Information

a) Creators:

Dr.  Maochao Xu


3. Purpose: Predict the last column. class: normal or anomaly

4. Number of instances:

Training data: 10, 246 observations (with 3522 anomalies)
Testing data: 14,946 observations (with 8221 anomalies)
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Answer the following questions. Note that models should be only trained on the training data!

a) *(5 Points) Using the bagging method to train your model, tuning your model if needed. Calculate the FP, FN, and sensitivity and specificity rates on the testing data.*

b) *(5 Points) Using the random forest method to train your model, tuning your model if needed. Calculate the FP, FN, and sensitivity and specificity rates on the testing data.*

c) *(20 Points) Report your findings based on those two methods. In your opinion, which rate(s) (FP/FN/sensitivity/specificity) is (are) more important? Are you satisfied with the important rates? If not, try a way to improve the rate(s), and explain your methods/results in detail.*

d) *(20 points) Use at least two Boost methods to train your model. Report the misclassification rate on the testing data. Calculate the FP, FN, and sensitivity and specificity rates on the testing data.*

e) *(10 points) How are your Boost methods compared to the RandomForest? Explain the possible reasons why it is better or worse.*

f) *(15 points) Select the top 5 important features from your best model. Now, remove those five important features from your data and rerun the best model. Report FP and FN. How would those five important features affect your FP or FN compared to the previous results? Explain your findings in detail.*

i) *(10 points) Summarize your results and explain any possible reason(s) that your best model can predict the data well. Are you satisfied with the FP or FN rate of your best model? If not, suggest a way to improve it, and report your results based on your idea.*

2. Use the AdaBoost.M1. algorithm to calculate the final classification error based on the following graph. The circled points are the misclassified observations. **Must show steps!** *(15 points)*

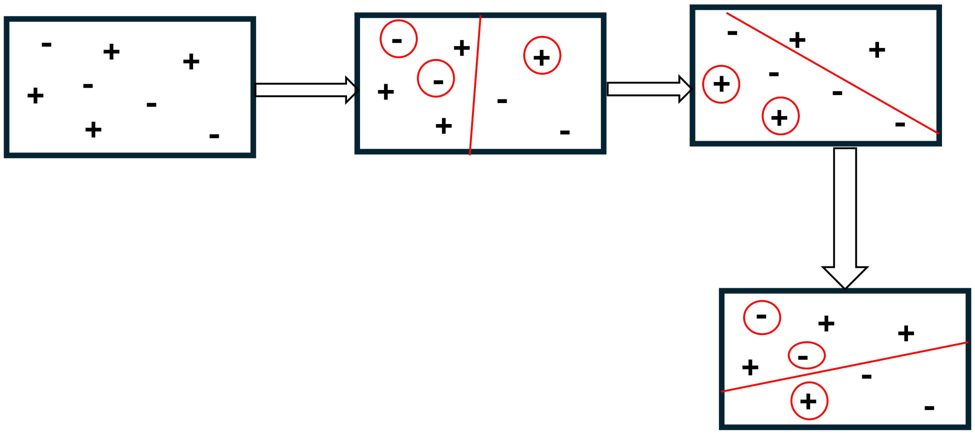

Figure 1: Classification based on reweighting.